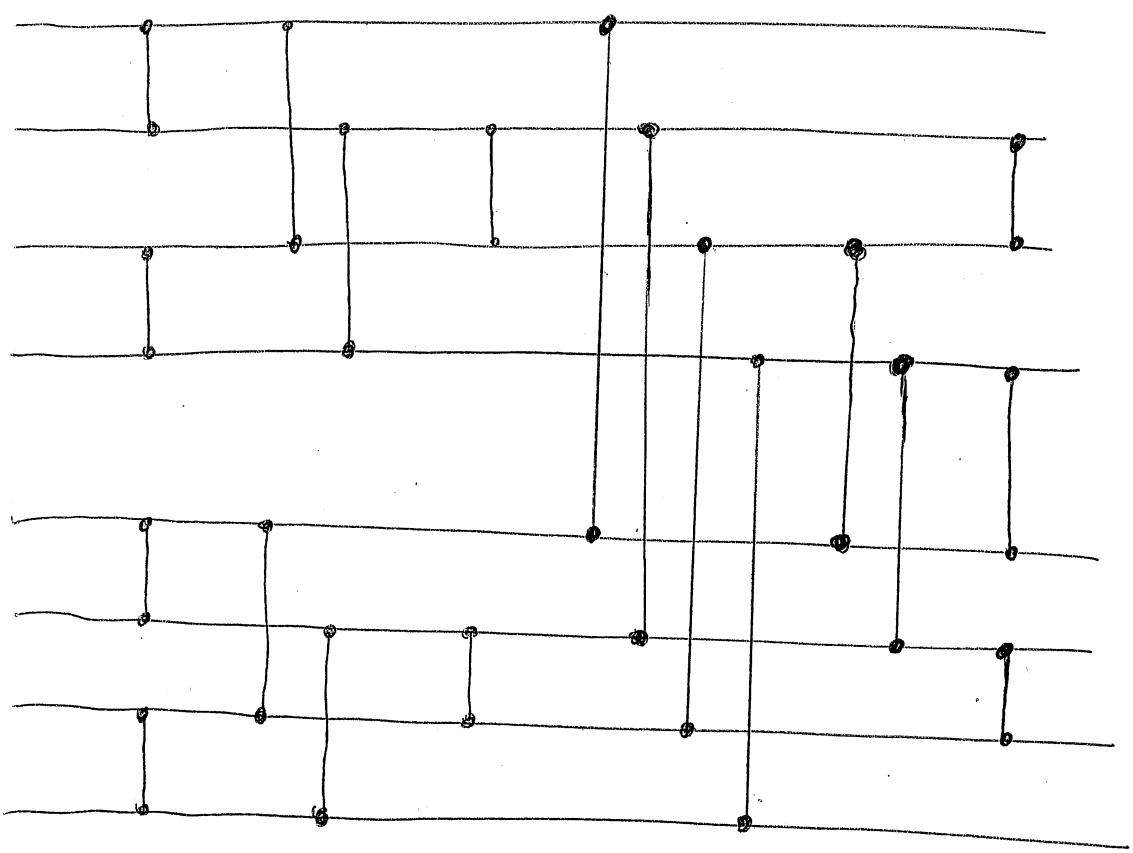
Lecture Merge Sort 02: other two algorithms that Sort a Single tile of data in parallel.
E.g. (We are focusing on sorting 256-element tile Here!)

Algorithms 1: odd-even Sorting Network (Batcher odd-even mergesort)

- Sorting Network: an abstract mathematical model of a network of wires and comparator modules that is used to sort a sequence of numbers.
  - differs from comparison-base sorting algorithms in that with sorting network the sequence of comparison is set in advance.

- A sorting network consists of two items: comparators and wires.



Lecture Merge Sort on GPU 02:

Batcher odd-even mergesort

We can observe that 1) horizontal lines are input wires, each of them carries a input number.

2) Vertical lines are comparators.

3) Clearly some comparator can do work in parallel.

4) You can use the algorithm to (build) output the (order) sequences of pairs of input numbers to be compared and swapped!

5) Very popular in parallel sorting!

6) Source code in python available on wikipedia for this Algorithm.

7) size: $O(n(\log n)^2)$, $\begin{cases} \text{depth } (\log n)^2 \\ n \text{ is input size.} \end{cases}$

depth: the maximum number of comparators along any path from input to an output.